Week 7 – Mini Project 1 (Non-Graded)

This is a practice mini project and is not graded, nor will it count towards program completion. However, we strongly recommend completing it to build your understanding of **probability concepts** and their application to real-world quality control scenarios.

This project focuses on applying basic probability and statistics to a manufacturing defect inspection dataset. You will calculate probabilities, conditional probabilities, and independence tests, along with visual analysis of defect patterns across shifts.

Case Study – Defect Inspection

You will be working with:

- **Mini_Project_1_DefectInspection.csv**

The objective of this activity is to analyze defect data from a manufacturing process, estimate probabilities of different defect types, and identify potential dependencies between shifts and defect occurrences.

How to approach this mini project:

1. **Load the dataset** into your preferred analytics tool (e.g., Python, Excel, Power BI).
2. **Review the structure of the dataset**, noting columns such as DefectA, DefectB, and Shift.
3. **Perform each calculation step-by-step**, interpreting what each probability represents.
4. **Use visuals** (such as bar charts or contingency tables) to understand patterns and dependencies.
5. **Reflect** on how observed data might differ from theoretical probabilities in practice.

Key tip:

👉 Focus on understanding *joint*, *marginal*, and *conditional probabilities*, as well as how to interpret independence in categorical data such as shifts and defect types.

Mini Project 1 – Defect Inspection: Questions

You are part of a **Quality Control team** tasked with analyzing inspection data from a manufacturing line. Use the dataset **Mini Project1_DefectInspection.csv** to answer the following questions:

1. **What is the probability of an item having Defect Type A?**
2. **What is the probability of an item having both Defect A and B?**
3. **What is** *P(Defect B | Defect A)* — the conditional probability of having Defect B given Defect A?
4. **How many items are defect-free** (no Defect A or B)?
5. **What proportion of defects occur in the Morning shift?**
6. **Create a contingency table** for Shift vs DefectA.
7. **Test independence:** Are defects and shift independent? Use proportions to assess.
8. **Estimate the joint probability** of A and B occurring together.
9. **Visualize shift-wise defect counts** using a bar chart.
10. **What is the probability of an item having at least one defect?**
11. **Estimate the expected defect count per item.**
12. **Compare marginal probabilities** of DefectA and DefectB.
13. **Estimate conditional probabilities:**
    - *P(DefectA | Shift = Morning)*
    - *P(DefectA | Shift = Evening)*
14. **What is P(no defects)?**
15. **Discuss:** What real-world factors could cause deviation from theoretical probabilities?